# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A solid sphere of radius $20$ cm is melted and recast into small solid spheres of radius $4$ cm. The number of small spheres is: 1

**(a)** $125$ **(b)** $130$ **(c)** $120$ **(d)** $150$

**2.** A cuboid $36$ cm $\times$ $24$ cm $\times$ $21$ cm is melted and recast into cubes of edge $3$ cm. The number of cubes is: 1

**(a)** $676$ **(b)** $680$ **(c)** $672$ **(d)** $674$

**3.** A cylinder of radius $3.5$ cm and height $6$ cm is melted and recast into cones of the same radius and height $1$ cm. The number of cones is: 1

**(a)** $24$ **(b)** $17$ **(c)** $19$ **(d)** $18$

**4.** Take $\pi = \frac{22}{7}$. The capacity of a hemispherical bowl of inner radius $21$ cm is: 1

**(a)** $23286$ cm$^3$ **(b)** $17463$ cm$^3$ **(c)** $19404$ cm$^3$ **(d)** $21345$ cm$^3$

**5.** Take $\pi = \frac{22}{7}$. The volume of a cone of radius $14$ cm and height $48$ cm is: 1

**(a)** $9856$ cm$^3$ **(b)** $8870$ cm$^3$ **(c)** $10842$ cm$^3$ **(d)** $11829$ cm$^3$

**6.** Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius $21$ cm and height $48$ cm is: 1

**(a)** $66528$ cm$^3$ **(b)** $59875$ cm$^3$ **(c)** $79835$ cm$^3$ **(d)** $73181$ cm$^3$

**7.** The diagonal of a cuboid measuring $4$ cm $\times$ $20$ cm $\times$ $22$ cm is: 1

**(a)** $29$ cm **(b)** $31$ cm **(c)** $32$ cm **(d)** $30$ cm

**8.** Take $\pi = \frac{22}{7}$. A toy is a cone of height $15$ cm placed on a hemisphere, both of radius $21$ cm. Its total volume is: 1

**(a)** $23700$ cm$^3$ **(b)** $31602$ cm$^3$ **(c)** $26334$ cm$^3$ **(d)** $28968$ cm$^3$

**9.** **Assertion (A):** The curved surface area of a cylinder is $2\pi rh$. 1
**Reason (R):** The total surface area of a cylinder is $2\pi rh + \pi r^2$.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The volume of a cone of base radius $r$ and height $h$ is $\pi r^2 h$. 1
**Reason (R):** The volume of a cone is one-third of the volume of a cylinder with the same base and height.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Take $\pi = \frac{22}{7}$. A toy is a cone of radius 10.5 cm and height 14 cm resting on a hemisphere of the same radius. Find its total volume and its total surface area (curved parts only). **2**

**12.** Take $\pi = \frac{22}{7}$. A cylinder of height 10 cm has volume 1540 $cm^3$. Find its radius and its curved surface area. **2**

**13.** A cuboid measures 6 cm × 6 cm × 7 cm. Find the length of its diagonal, its total surface area and its volume. **2**

**14.** Take $\pi = \frac{22}{7}$. A toy is a cone of radius 31.5 cm and height 42 cm resting on a hemisphere of the same radius. Find its total volume and its total surface area (curved parts only). **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A solid sphere of radius 9 cm is melted and recast into a solid cylinder of radius 6 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). **3**

**16.** Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 14 cm and height 30 cm is poured into an empty rectangular tank with a base 44 cm × 21 cm. Find the height of water in the rectangular tank. **3**

**17.** Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 10.5 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 4851 $cm^3$ of soup. **3**

**OR**

A solid sphere of radius 9 cm is melted and recast into a solid cylinder of radius 2 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). **3**

**18.** Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 21 cm and height 20 cm is poured into an empty rectangular tank with a base 22 cm × 21 cm. Find the height of water in the rectangular tank. **3**

**OR**

Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 10.5 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 12127.5 $cm^3$ of soup. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A cylindrical water bottle has a base radius of 7 cm and a height of 20 cm. Take $\pi = \frac{22}{7}$. **5**

(i) Find its volume.
(ii) Find its curved surface area.
(iii) Find its total surface area.
(iv) How many such bottles are needed to fill a bucket of capacity 30.8 litres?

**20.** Take $\pi = \frac{22}{7}$. An ice-cream cone of radius 21 cm and height 28 cm is completely filled and topped with a hemisphere of ice-cream of the same radius. Find the total volume of ice-cream and the volume needed for 20 such cones. **5**

**OR**

A solid sphere of radius 9 cm is melted and 5 identical solid cylinders of radius 1 cm are made from the metal. Find the height of each cylinder. **5**

## END OF QUESTION PAPER — Total: 40 Marks